MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: GAMAT301
Course Name: MATHEMATICS FOR COMPUTER AND INFORMATION SCIENCE – 3

Max. Marks: 60 | Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		The p.m.f. of the amount of memory X (GB) in a purchased flash drive is given. x: 1, 2, 4, 8, 16 P(x): .05, .10, .35, .40, .10 Find the mean and Variance		(3)
2		8 Coins are tossed 256 times. In how many tosses do you expect no heads ?		(3)
3		The mileage which car owners get with a certain kind of radial tire is a random variable having an exponential distribution with mean 40000 km. Find the probability that one of these tires will last at least 20000 km.		(3)
4		If X follows $N(3,2)$, find the value of k such that $P(\|X-3\| > k) = 0.05$		(3)
5		Suppose we know that the number of items produced in a factory during a week is a random variable with mean 500. If the variance of a week's production is known to equal 100, then what can be said about the probability that this week's production will be between 400 and 600?		(3)
6		Classify stochastic processes with examples.		(3)
7		Whether or not it rains follows a 2 state Markov chain. If it rains one day, then it will rain the next with probability 1/2 or will be dry with probability 1/2. Overall, 40 percent of days are rainy. If it is raining on Monday, find the probability that it will rain on Thursday.		(3)
8		J plays a new game every day. If J wins a game, then she wins the next one with probability 0.6; if she has lost the last game but won the one preceding it, then she wins the next with probability 0.7; if she has lost the last 2 games, then she wins the next with probability 0.2. What proportion of games does J win?		(3)

PART B				
Answer any one full question from each module. Each question carries 9 marks				
Module 1				
9	a)	Suppose that the number of drivers who travel between a particular origin and destination during a designated time period has a Poisson distribution with parameter $\mu = 20$. What is the probability that the number of drivers will be at most 3 ?		(4)
	b)	If the joint probability density function of random variables of X and Y is given by $f(x, y) = c(2x + 3y),\ x = 0,1,2;\ \ y = 1,2,3$. Find (i) the value of c (ii) the marginal probability distributions of X and Y (iii) the probability distribution of $X + Y$.		(5)
10	a)	If 6 of 18 new buildings in a city violate the building code, what is the probability that a building inspector, who randomly selects 4 of the new buildings, will catch (i) none (ii) exactly one of the new buildings that violate the building code?		(4)
	b)	A random variable X takes values -3, -2, -1, 0, 1, 2, 3 such that $P(X = 0) = P(X > 0) = P(X < 0)$ and $P(X = -3) = P(X = -2) = P(X = -1) = P(X = 1) = P(X = 2) = P(X = 3)$. Obtain the probability mass function and distribution function of X.		(5)
Module 2				
11	a)	Buses arrive at a specified stop at 15 min intervals starting at 7 am, that is they arrive at 7, 7:15, 7:30, 7:45 and so on. If a passenger arrives at the stop at a random time that is uniformly distributed between 7 and 7:30 am, find the probability that he waits (i) less than 5 min for a bus (ii) at least 12 min for a bus.		(4)
	b)	The joint pdf of a two-dimensional random variable (X, Y) is given by $$f(x,y) = \begin{cases} k(6 - x - y); & 0 < x < 2, 2 < y < 4 \\ 0; & elsewhere \end{cases}$$ Find (i) the value of k, (ii) $P(X < 1, Y < 3)$, (iii) $P(X + Y < 3)$		(5)
12	a)	Let X follows exponential distribution with pdf $$f(x) = \begin{cases} \frac{1}{5} e^{-x/5}, & x > 0 \\ 0, & otherwise \end{cases}$$ Evaluate (i) $E(X)$ (ii) $E(Y)$ (iii) $P(X + Y \geq 1)$		(4)

	b)	Suppose the Rockwell hardness of a particular alloy is normally distributed with mean 70 and standard deviation 3. If a specimen is acceptable only if its hardness is between 67 and 75, what is the probability that a randomly chosen specimen has an acceptable hardness?		(5)
		Module 3		
13	a)	Suppose that people immigrate into a territory according to a Poisson process with rate $\lambda = 2$ per day. **(a)** Find the probability there are 10 arrivals in the following week (of 7 days). **(b)** Find the expected number of days until there have been 20 arrivals.		(5)
	b)	The lifetime of a special type of battery is a random variable with mean 40 hours and standard deviation 20 hours. A battery is used until it fails, at which point it is replaced by a new one. Assuming a stockpile of 25 such batteries, the lifetimes of which are independent, approximate the probability that over 1100 hours of use can be obtained.		(4)
14	a)	If people arrive at a book stall in accordance with a Poisson process with mean rate of 3 per minute, find the probability that the interval between 2 consecutive arrivals is (i) more than 1 minute (ii) between 1 minute and 2 minutes (iii) 4 minutes or less.		(6)
	b)	Let $X_1, X_2, \dots,, X_{10}$ be independent Poisson random variables with mean 1. Use the Markov inequality to get a bound on $P\{X_1, X_2, \dots,, X_{10} \geq 15\}$.		(3)
		Module 4		
15		The TPM of a Markov chain $\{X_{n,} n \geq 0\}$ having 3 states 1,2 and 3 is $\begin{matrix} 0.2 & 0.3 & 0.5 \\ 0.1 & 0.6 & 0.3 \\ 0.4 & 0.3 & 0.3 \end{matrix}$ and the initial probability distribution is $P(0) = [0.5\ \ 0.3\ \ 0.3]$. Find *(i)* $P\{X_2 = 2\}$ *(ii)* $P\{X_3 = 3,\ X_2 = 2,\ X_1 = 1, X_0 = 3\}$ *(iii)* $p^{(2)}$, the 2 step TPM		(9)

16		Capa plays either one or two chess games every day, with the number of games that she plays on successive days being a Markov chain with transition probabilities $P_{1,1} = 0.2$, $P_{1,2} = 0.8$, $P_{2,1} = 0.4$, $P_{2,2} = 0.6$ Capa wins each game with probability p. Suppose she plays two games on Monday. **(a)** What is the probability that she wins all the games she plays on Tuesday? **(b)** What is the expected number of games that she plays on Wednesday? **(c)** In the long run, on what proportion of days does Capa win all her games.		(9)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
THIRD SEMESTER B. TECH DEGREE EXAMINATION, NOVEMBER 2025

Course Code: PCITT302

Course Name: Computer Organisation and Architecture

Max. Marks: 60		Duration: 2 hours 30 minutes

PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Why are fixed-length instructions preferred in RISC architectures? Explain.	CO1	(3)
2		Discuss the different phases of an instruction execution.	CO1	(3)
3		Discuss about structural hazard and control hazard that occur in RISC instruction pipeline.	CO2	(3)
4		Consider the following instruction sequence: BEQ R1, R2, L1 ADD R3, R4, R5 L1: SUB R6, R7, R8 Assume that the processor uses delayed branching with one delay slot. Reorder or modify the instruction sequence using delayed branching so that the pipeline efficiency is improved.	CO2	(3)
5		A set-associative cache consists of 64 lines, or slots, divided into four-line sets. Main memory contains 4K blocks of 128 words each. Show the format of main memory addresses.	CO3	(3)
6		Discuss about the DRAM subsystem organization.	CO3	(3)
7		What is multithreading in processor design?	CO4	(3)
8		Differentiate between SISD and SIMD architectures with examples.	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

Module 1

9	a)	Write an assembly language program to derive the expression X = A*B+C*D a) in a stack-based computer with zero address instructions b) In an accumulator-based CPU with one address instructions c) In register-based CPU with two address instructions d) In register-based CPU with three address instructions	CO1	(5)
	b)	Consider a computer that has a byte addressable memory organized in 32-bit words according to the big-endian scheme. A program reads ASCII characters entered at a keyboard and stores them in successive byte locations, starting at location 1000. Show the contents of the two memory words starting at location 1000 after the word "MOUNTAINS" has been entered.	CO1	(4)

10	a)	Registers R1 and R2 of a computer contains the decimal values 1200 and 4600. What is the effective address of the memory operand in each of the following instructions? (a) LW 20(R1), R5 (b) MOV #3000, R5 (c) SW R5,30(R1, R2) (d) ADD -(R2), R5 (e) SUB (R1) +, R5	CO1	(5)
	b)	The memory unit of a computer has 256K words of 32 bits each. The computer has an instruction format with four fields: an operation code field, a mode field to specify one of seven addressing modes, a register address field to specify one of 60 processor registers, and a memory address. Specify the instruction format and the number of bits in each field if the in instruction is in one memory word.	CO1	(4)
		Module 2		

11	a)	A 5-stage pipelined processor has the stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory access (MEM) and Write Back (WB). Each stage requires only one cycle. Operand forwarding is used in the pipeline. What is the number of clock cycles needed to execute the following sequence of instructions?
		//ADD TWO INTEGER ARRAYS
		LW R4, # 400
		LW R1, 0 (R4) ; Load first operand
		LW R2, 400 (R4) ; Load second operand
		ADDI R3, R1, R2 ; Add operands
		SW R3, 0 (R4) ; Store result
		SUB R4, R4, #4 ; Calculate address of next element
			CO2	(5)

	b)	Assume a pipelined processor with a single memory unit shared for both instruction fetch and data access. Show how a structural hazard arises when executing a LW followed by a SW, and propose an architectural solution to eliminate this hazard.	CO2	(4)
12	a)	Consider an instruction pipeline with four stages with the stage delays 5 nsec, 6 nsec, 11 nsec, and 8 nsec respectively. The delay of an inter-stage register stage of the pipeline is 1 nsec. What is the approximate speedup of the pipeline in the steady state under ideal conditions as compared to the corresponding non-pipelined implementation?	CO2	(5)
	b)	Discuss various types data hazards in a RISC Instruction pipeline with appropriate examples.	CO2	(4)
		Module 3		
13	a)	Consider a memory system that uses a 32-bit address to address at the byte level, plus a cache that uses a 64-byte line size. Assume a four-way set-associative cache with a tag field in the address of 9 bits. Show the address format and determine the following parameters: number of addressable units, number of blocks in main memory, number of lines in set, number of sets in cache, number of lines in cache, size of tag.	CO3	(5)
	b)	Explain DRAM scheduling policies.	CO3	(4)
14	a)	Given a direct-mapped cache, initially empty, and the following memory access pattern (all byte addresses/accesses, 32-bit addresses with block size of	CO3	(3)

		4B) 8, 0, 5, 32, 0, 42, 9. What kinds are the different cache misses, and what blocks are in the cache after these accesses if the cache has a capacity of 16B?		
	b)	A cache has a 95% hit ratio, an access time of 100ns on a cache hit, and an access time of 800ns on a cache miss. Compute the effective access time.	CO3	(3)
	c)	Differentiate between open-page and close-page row buffer management policies in DRAM.	CO3	(3)
Module 4				
15	a)	Describe the role of vector hardware in parallel processing. How does it improve performance?	CO4	(5)
	b)	Discuss the architectural features of Intel Core i7 that support parallel processing.	CO4	(4)
16	a)	Differentiate between SISD, SIMD, and MIMD architectures with suitable examples.	CO4	(5)
	b)	Explain how shared memory multiprocessors manage data consistency among cores.	CO4	(4)
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MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PCITT303
Course Name: DATA STRUCTURES

Max. Marks: 60 | Duration: 2 hours 30 minutes

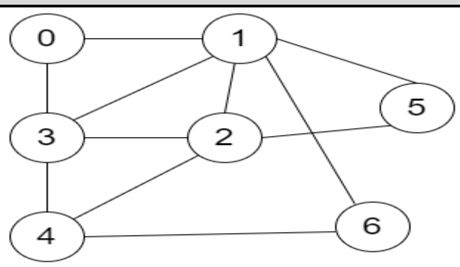
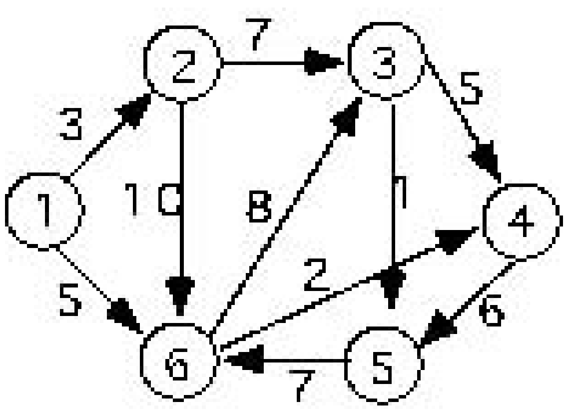
PART A

		Answer all questions. Each question carries 3 marks	CO	Marks
1		Define Abstract Data Type (ADT) and differentiate it from Concrete Data Type (CDT) with examples.	CO1	(3)
2		Represent a two-dimensional array in memory. Explain row-major and column-major representations.	CO1	(3)
3		Explain how binary search improves efficiency over linear search.	CO2	(3)
4		What is the load factor in hashing? How does it impact performance?	CO2	(3)
5		Compare array and linked list in terms of memory usage and insertion/deletion complexity.	CO3	(3)
6		List two real-life applications of doubly linked lists	CO3	(3)
7		Define complete, perfect, and full binary trees with suitable examples.	CO4	(3)
8		Represent a graph using both adjacency list and adjacency matrix	CO4	(3)

PART B

Answer any one full question from each module. Each question carries 9 marks

		Module 1		
9	a)	Explain time and space complexity of an algorithm using tabular and graphical illustrations.	CO1	(5)
	b)	Write an algorithm to implement a circular queue using array.	CO1	(4)
10	a)	Write an algorithm for stack operations (PUSH and POP) using arrays.	CO1	(5)
	b)	Compare priority queues and deques in terms of structure, use cases, and operations.	CO1	(4)
		Module 2		
11	a)	Evaluate the given expression using a stack. 3-(4/2) + (1*5) + 6	CO2	(5)
	b)	Sort the array [29, 10, 14, 37, 13] using merge sort and explain each step	CO2	(4)

12	a)	Construct a hash table for the elements [23, 43, 13, 37, 33] using the division method with table size 7. h1(k)=k%7. Use double hashing for collision resolution techniques with secondary hash function h2(k)=5−(k%5).	CO2	(5)
	b)	Compare bubble sort and insertion sort with respect to time complexity and use cases.	CO2	(4)
Module 3				
13	a)	Write an algorithm to insert a node at the beginning and end of a singly linked list.	CO3	(5)
	b)	Write a function to search for an element in a singly linked list.	CO3	(4)
14	a)	Write a C program for polynomial addition using linked lists.	CO3	(5)
	b)	Show how queue can be implemented using linked list. Write the algorithm for enqueue operation	CO3	(4)
Module 4				
15	a)	 Apply BFS and DFS on the above graph and show the final traversal order	CO4	(5)
	b)	Construct an expression tree for the expression (a+b*c) + ((d*e+f)*g)	CO4	(4)
16	a)	Apply Dijkstra's algorithm to find the shortest path from vertex 1 in the given graph: 	CO4	(5)
	b)	Construct a Binary Search Tree (BST) with values [50, 30, 70, 20, 40, 60, 80] and perform in-order traversal.	CO4	(4)

MODEL QUESTION PAPER

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR

Course Code: PBITT304
Course Name: Database Management System

Max. Marks: 40 | Duration: 2 hours 30 minutes

PART A

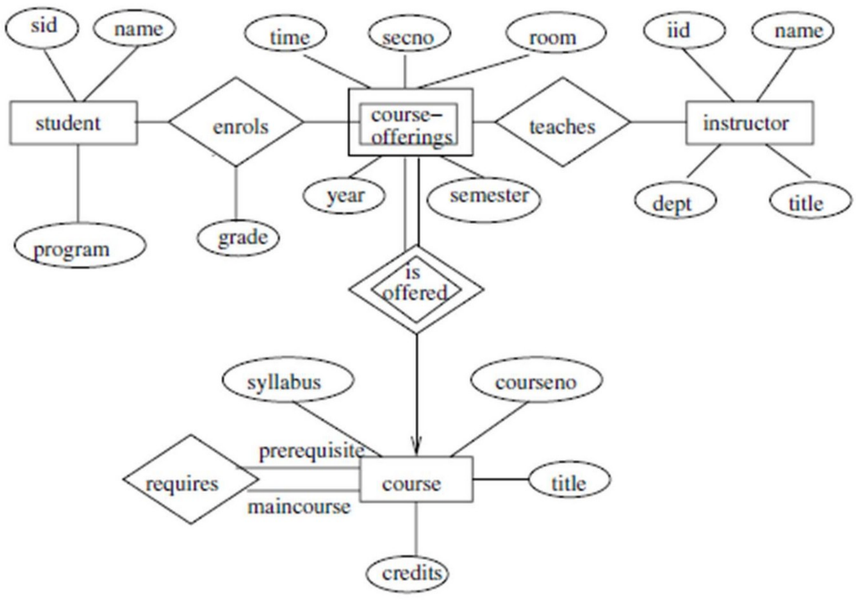
		Answer all questions. Each question carries 2 marks.	CO	Marks
1		Distinguish between total and partial participation constraints in an ER diagram with the help of an example.	1	(2)
2		List any two categories of database users, highlighting the important characteristicsof each category.	1	(2)
3		Illustrate the different types of SQL Joins with examples.	2	(2)
4		Given a relation R(A,B,C,D,E,F) with functional dependencies $A \rightarrow B$, $B \rightarrow D$, $D \rightarrow EF$, $F \rightarrow A$, compute {D}+ and {EF}+.	2	(2)
5		What is log-based recovery in a database system?	3	(2)
6		What is conflict-serializability? How can you check whether a given schedule is conflict serializable or not?	3	(2)
7		How can ODBC API be used to access data stored in various databases from an application?	4	(2)
8		List the types of database testing.	4	(2)

PART B

Answer any one full question from each module. Each question carries 6 marks.

Module 1

			CO	Marks
9	a)	A database is being constructed to keep track of the teams and games of a football league. A team has a number of players. For the team, we are interested in storing the team id, team name, address, date established, name of the manager, and name of the coach. For the player, we will store the player ID in the team, date of birth, date joined, position, etc. Each team plays games against the other team in a round-robin fashion. For each game, we will store game ID, date held, score, and attendance (an attribute to designate whether the participating teams have attended the game). Games are generally taking place at various stadiums in the country. For each stadium, we will keep its size, name, and location. Develop a complete E-R diagram (including cardinalities) for the given scenario. Make reasonable assumptions during your development phases, if needed and state them clearly.	1	(4)
	b)	Distinguish between weak entity and strong entity with the help of an	1	(2)

		example.		
10	a)	Develop a relational model from the given ER diagram. Clearly mention the primary key of each table. 	1	(4)
	b)	Explain the three-schema architecture of a database management system with the help of a neat diagram.	1	(2)
		Module 2		
11	a)	Consider the following relations which are part of the University database. Instructor (id, name, dept_name, salary) Course (course_id, title, dept_name, credits) Section (course_id,sec_id,semester,year,building,room_number,time_slot_id). Give SQL queries to answer the following questions: 1. Find the titles of courses in the IT department that have 4 credits. 2. Find all instructors earning the highest salary. 3. Increase the salary of each instructor in the IT department by 10%. 4. Delete all courses that have never been offered.	2	(4)
	b)	Write SQL queries to create a view, update the view and drop the view.	2	(2)
12	a)	Given a relation R(A, B, C, D) and the set of functional dependencies FD = {AB->CD, B->C}.Determine whether R is in 2NF or not. If not, convert into 2NF.	2	(4)
	b)	Explain insertion, updation and deletion anomalies in relational database design using suitable examples.	2	(2)
		Module 3		
13	a)	Consider a database transaction to transfer 'x' dollars from Account A to Account B. Illustrate the need to ensure all the ACID properties in a	3	(4)

		transaction with this example.		
	b)	Explain two phase locking protocols in concurrency control.	3	(2)
14	a)	Describe the recovery techniques based on Immediate Update and Deferred Update with an example.	3	(4)
	b)	Explain the different states of a database transaction.	3	(2)
Module 4				
15	a)	Differentiate between user-interface testing and data testing.	4	(6)
16	a)	Explain the architecture of an ODBC API, clearly mentioning the steps in connecting an application to a back-end database.	4	(6)
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